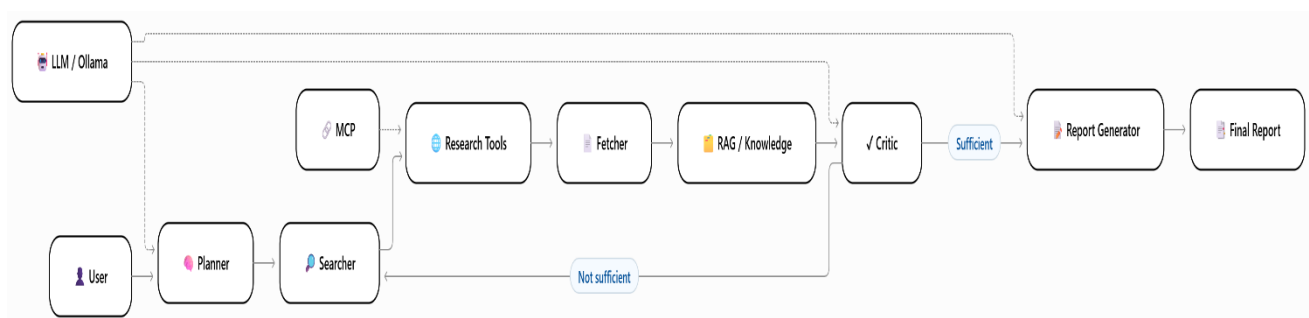


SYSTEM ARCHITECTURE DESCRIPTION

The Agentic Deep Researcher uses a multi-agent architecture in which each agent is responsible for a specific stage of the research process.

The system starts when the User enters a research question through the interface. The Planner Agent analyzes the question and breaks it into smaller sub-questions, creating a structured research plan. The Searcher Agent uses this plan to find relevant information from web and academic sources such as Tavily, Exa, and Semantic Scholar. The Fetcher Agent then retrieves the useful content from the selected sources. The retrieved information is processed and stored using RAG and a knowledge store such as ChromaDB, allowing relevant information to be retrieved during later stages. The Critic Agent examines the collected information to check whether it is relevant, sufficient, and covers the important aspects of the research question. If gaps are found, the system sends the task back to the Searcher Agent for additional research. This iterative process continues until sufficient information is available. Finally, the Report Generator uses the verified research information to prepare a structured final report. The report can include findings, source references, inline citations, and confidence information. The overall workflow can be managed using LangGraph or CrewAI, while MCP provides a standardized connection between the AI agents and external research tools. An LLM/Ollama provides the reasoning and language-generation capabilities required by the agents.



MODULES DESCRIPTION

User Interface Module : This module provides the interface through which the user enters a research question and receives the final research report. It acts as the primary interaction point between the user and the system.

Planner Agent Module : The Planner Agent analyzes the user's research question and breaks it into smaller, meaningful sub-questions. It creates a structured research plan and determines the areas that need to be investigated.

Searcher Agent Module : The Searcher Agent converts the research plan into suitable search queries and searches for relevant information. It can interact with tools such as Tavily, Exa, and Semantic Scholar to obtain web-based and academic sources.

Fetcher Agent Module : The Fetcher Agent retrieves the actual content from the sources identified by the Searcher Agent. It extracts useful information and handles unsuccessful or inaccessible sources where possible.

RAG and Knowledge Management Module : This module manages the retrieved research information. Using Retrieval-Augmented Generation (RAG) and a knowledge store such as ChromaDB, relevant information can be stored, organized, and retrieved when required.

Critic Agent Module : The Critic Agent evaluates the collected information for relevance, completeness, and research gaps. If important information is missing, it can request additional research, creating an iterative improvement loop.

Agent Orchestration Module : This module manages the sequence and communication between the different agents. Frameworks such as LangGraph , CrewAI can be used to control the workflow and ensure that each agent performs its assigned task.

MCP and Tool Integration Module : The Model Context Protocol (MCP) module provides a standardized mechanism for connecting AI agents with external tools. It enables the system to access search and research services without tightly coupling each agent to individual tools.

LLM Processing Module : The LLM acts as the reasoning and language-processing component of the system. It supports question analysis, planning, query generation, evaluation, and report generation. Ollama can be used to run a compatible model locally.

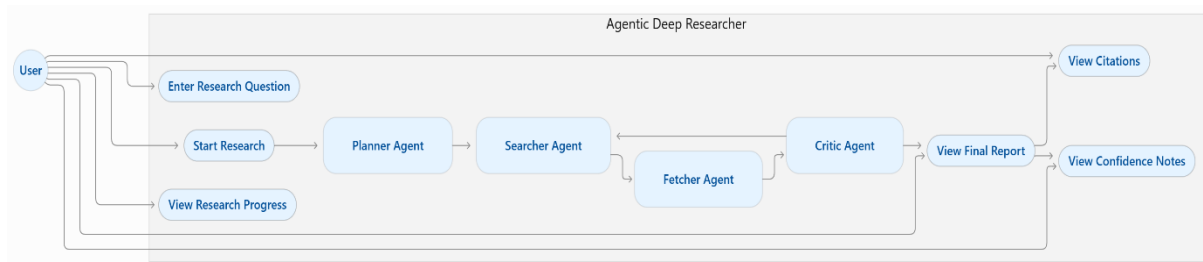
Report Generation Module : This module combines the verified research information and generates the final structured report. The output can contain findings, source references, inline citations, and confidence information.

Error Handling and Validation Module : This module handles issues such as failed searches, inaccessible webpages, incomplete source content, and invalid results. It helps the system continue the research process without stopping unnecessarily.

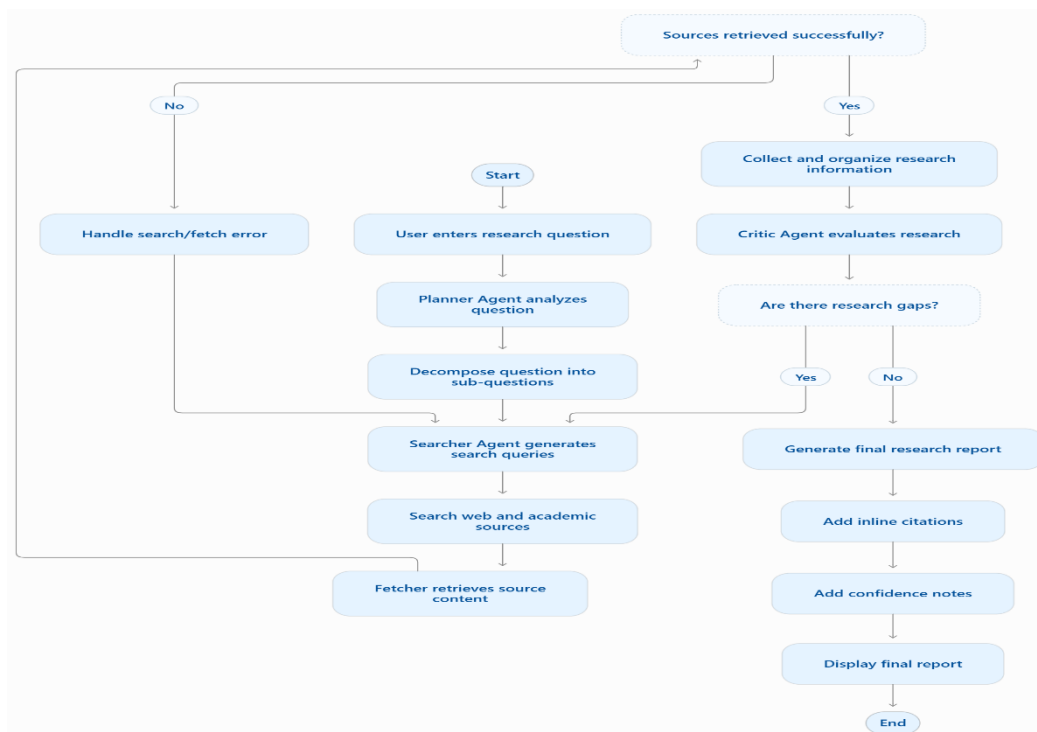
Overall, these modules work together to provide a structured research workflow from question analysis and information retrieval to evaluation and final report generation.

UML DIAGRAMS

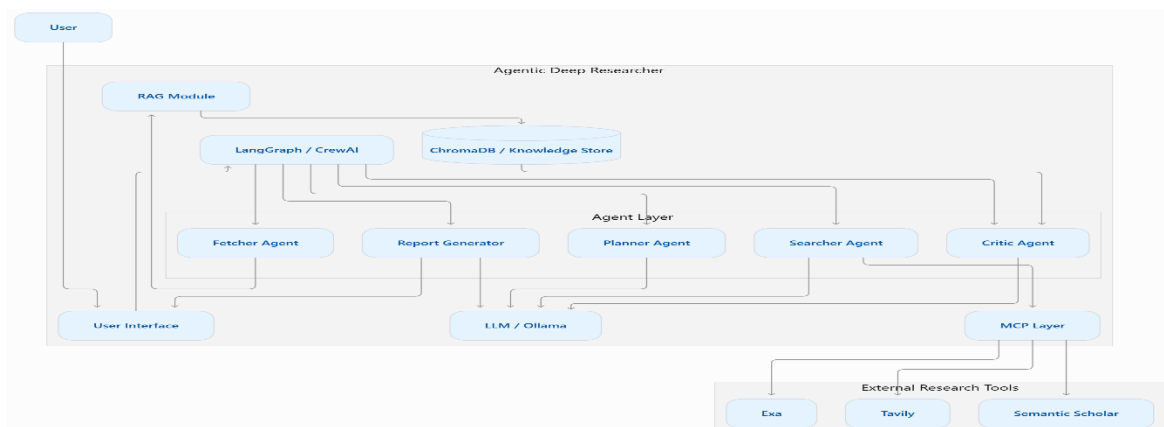
Use Case Diagram :



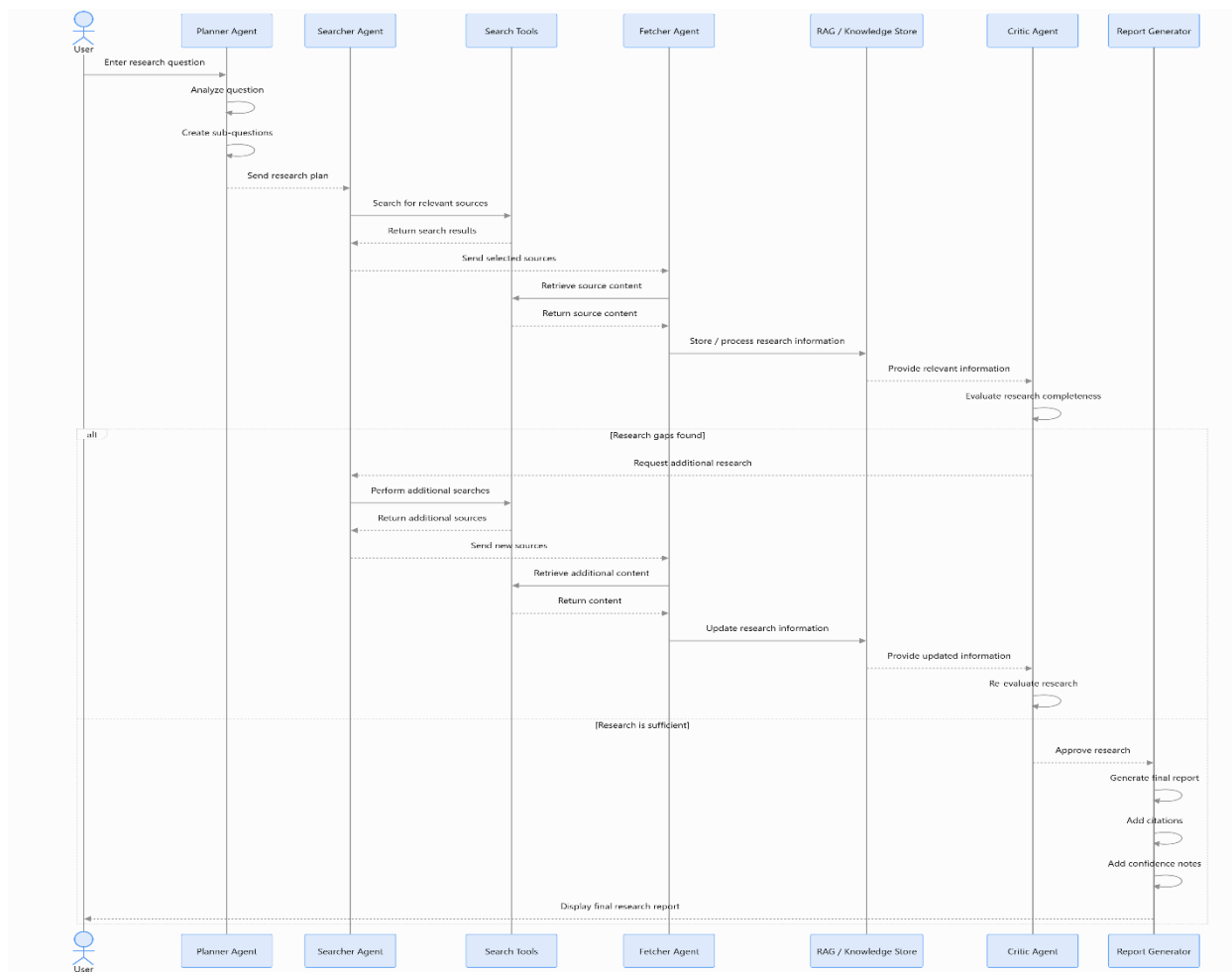
Activity Diagram :



Component Diagram :



Sequence Diagram :



Class Diagram :

